

Chapter 4. Lagrange's Theorem

4.1 Cosets



In this video, we will introduce the main objects needed to prove Lagrange's Theorem.

What are cosets?

Definition 4.1. Let H be a subgroup of a group G and $a \in G$. The set $Ha = \{ha \mid h \in H\}$ is called the **right coset** of H in G **generated** by a , while $aH = \{ah \mid h \in H\}$ is called the **left coset** of H in G **generated** by a .

A couple of remarks:

①

Observe that in **additive notation** the left and right cosets are:

$$H + a = \{h + a \mid h \in H\}, \quad a + H = \{a + h \mid h \in H\}$$

②

It is important to notice that **cosets in a group G is a subset of G .**

Cosets are not elements in G !

Example 4.2. Find all the left and right cosets of $H = \{\varepsilon, (1\ 3)\}$ in S_3 .

Recall that $S_3 = \{\varepsilon, (1\ 2), (1\ 3), (2\ 3), (1\ 2\ 3), (1\ 3\ 2)\}$.

It is clear $\varepsilon H = H = H\varepsilon$

$$(1\ 2)H = ? \quad H(1\ 2) = ?$$

Left coset of
H generated by (1 2)

right coset of H
generated by (1 2)

By definition we have

$$(1\ 2)H = \{(1\ 2)\varepsilon, (1\ 2)(1\ 3)\} = \\ = \{(1\ 2), (1\ 3\ 2)\}$$

$$(1\ 2)(1\ 3) = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix} = (1\ 3\ 2)$$

$$H(1\ 2) = \{\varepsilon(1\ 2), \underbrace{(1\ 3)(1\ 2)}\} = \{(1\ 2), (1\ 2\ 3)\}$$

$$(1\ 3)(1\ 2) = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix} = (1\ 2\ 3)$$

$$\begin{aligned} \underline{(1\ 3)H} &= \left\{ (1\ 3)\varepsilon, \underline{(1\ 3)(1\ 3)} \right\} = \\ &= \left\{ (1\ 3), \varepsilon \right\} = \underline{H} \end{aligned}$$

$$\begin{aligned} (1\ 3\ 2)H &= \left\{ (1\ 3\ 2)\varepsilon, \underline{(1\ 3\ 2)(1\ 3)} \right\} \\ &= \left\{ (1\ 3\ 2), (1\ 2) \right\} = (1\ 2)H \end{aligned}$$

$$(1\ 3\ 2)(1\ 3) = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix} = (1\ 2)$$

Example 4.3. Find all the cosets of the subgroup $4\mathbb{Z}$ in the additive group $\mathbb{Z}$. In this case, since $\mathbb{Z}$ is an abelian group, it is enough to compute the right (or the left) cosets.

For $a \in \mathbb{Z}$, we want to compute:

$$4\mathbb{Z} + a = a + 4\mathbb{Z} = \bar{a} \text{ in } \mathbb{Z}_4$$

We only have four cosets, which are the elements of $\mathbb{Z}_4$:

$$4\mathbb{Z} + 0 = \{ \dots, -12, -8, -4, 0, 4, 8, 12, \dots \}$$

$$4\mathbb{Z} + 1 = \{ \dots, -11, -7, -3, 1, 5, 9, 13, \dots \}$$

$$4\mathbb{Z} + 2 = \{ \dots, -10, -6, -2, 0, 2, 6, 10, \dots \}$$

$$4\mathbb{Z} + 3 = \dots$$